Lab III

Relational Operators, Logical Operators and Conditionals

Asad Ullah

Electrical Ai

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Section B

Reg No: BF25NWELE0686

Learning objectives:

The following types will be covered in this lab session

- Comparison/Relational Operators
- Logical Operators
- If Statement
- If - else statement
- else - if Statement

➯ Theory: Comparison/Relational Operators

Operator Name	Syntax
Less than	a<b
Less than equal to	a<=b
Greater than	a>b
Greater than equal to	a>=b
Equal to	a==b
Not equal to	a !=b

➯ Logical operators:

Operator Name	Syntax
Logic Negation (not)	!a
Logic And	a && B
Logic or	a || b

➯ Decision Control Structures:

- Till now we have used sequence control structure in the programs, in which the various steps are executed sequentially i.e. in the same order in which they appear in the program.
- In C++ programming the instructions are executed sequentially, by default. At times, we need a set of instructions to be executed Depending upon different conditions to be satisfied.
- In such cases we have to use decision control instructions. This can be achieved in C using;
    - The if statement
    - The if-else statement
    - The else - if statement
    -

The C/C++ "if" Statement

- The **if** statement controls conditional branching. The body of an if statement is executed if the value of the expression/condition specified in the if statement is true. The syntax of the if statement is as follows:

Syntax:

```
If (expression)  {
   Block of statement;
 }
```

Example:

- Write a program in which it takes a number from keyboard as an input and if the number is greater than 100 it prints "The number is greater than hundred".

```
#include <iostream>
#include<conio.h>

int main()
{
      int number ;
      cout<< "Enter an integer\n";
      cin>> number;
      if ( number >100 )
            cout<<"The number is greater than 100";
      getch();

}
```

- Note we did not use curly brackets '{}'for body of if **if ()** because it did not include multiple statements (block of statements). If it includes multiple statements then it would have been something like this.

```
If(number<100){
cout<< "The number is greater than 100\n";
cout<< "No doubt that the number is greater than 10
}
```

The C++ if-else Statement:

- **if - else** statement is similar to **if** with the addition of else statement. If the condition is false in **if** then its body will be skipped and the else statement's body will be executed.

Syntax:

```
if (expression)
{
    Block of statement;
}
else
{
    Block of statement;
}
```

Example:

- Write a program in which it takes two numbers from keyboard as input and subtract larger number from smaller.

```
#include <iostream>
#include<conio.h>


int main()
{
        int a,b ;

        cout << "Enter first number\n";
        cin >> a;
        cout <<  "Enter second number\n";
        cin >> b;

        if ( a >=b )
        // this condition can also be written as if(a>b || a==b)
                cout <<a <<"-" <<b <<"=" <<a-b;
        else
                cout <<b <<"-" <<a << "=" <<b-a;

        getch();

}
```

Nested if-else statement:

- **Nested if is basically if inside if.**

Example:

- Write a program which take a number from keyboard and checks the number whether that number is less than 100 or not if that number is less than 100 than check that is it less than 50 or not.

```
#include <iostream>
#include <conio.h>

int main()
{
        int number ;
        cout << "Enter an integer\n";
        cin >>number;
        if ( number <100 )
        {
                cout<< "Yes the number is less than 100";
                if ( number <50)
                {
                        cout<< " and number is also less than 50";
                }
                else
                {
                        cout<< " but the number is not less than 50";
                }
        }
        else
                cout<< "No the number is not less than 100";
        getch();

}
```

✦ The above program checks if the number is less than 100. This check is done in the first **if** statement.

- If the result is true the program enters the second **if** statement which is also called the **nested if** statement. Here another check is performed. The number is checked if it less than 50 or not. If the number is less than 50 it is conveyed to the user. The rest is pretty self-explanatory.

The C++ else-if Statement:

- Sometimes we wish to make a multi-way decision based on several conditions. The most general way of doing this is by using the else if variant on the if statement. This works by cascading several comparisons. As soon as one of these gives a true result, the following statement or block is executed, and no further comparisons are performed.

```
Syntax:
if(expression)
{
Block of statement;
} else
if(exprerssi
on)
{
Block of statement;
} else
if(exprerssi
on)
{
Block of statement;
}
. . // you can add as many else-if statements as many you need
. .
. .
else
{
Block of statement;
}
```

Lab Tasks:

1. Write a C program that prompts the user to enter two numbers.
2. Use comparison operators to check if the first number is equal to, greater than, or less than the second number.
3. Display appropriate messages based on the comparison results.

Task 1: Comparison / Relational Operators

```
#include <iostream>                 // tells the cmpiler to include standard
                                       ninput output library
using namespace std;                // allows us to use cout and
                                       cin without tyiping std:: for every one
int main(){                         //starting point of the main function
int a,b;                            // variable declaration
cout<<"enter an integer ";          // prompt for user to input data
cin>>a;                             // takes input from user and places it in
                                       the variable
cout<<"enter 2nd integer ";         // 2nd prompt for user to input data
cin>>b;                             // takes data from user and places it
                                       in the variable
if(a>=b){                           // condition for true
 cout<<a<<"is greater than or equal to  "<<b;}   //  prints the result if the
                                                     if condition met
else                                             // part which runs if the if
                                                    condition fails
 cout<<a<<"is less than "<<b;                    //prints result for a
                                                   smaller than b
return 0;                           // tells the operating system that the
                                       program finished successfully
}
```

Explanation

This program asks the user to enter two numbers.
The relational operators **==, >, and** < are used to compare the numbers.

Depending on the comparison result, the program displays an appropriate message indicating whether the numbers are **equal, greater, or smaller**.

Task 2: Logical Operators

Program

```
#include <iostream>                   // tells the compiler to include standard
                                         input output library
using namespace std;                  // allows us to use cout without having
                                         to type std:: for each

int main(){                           // the starting point from where the
                                         computer begins executing program

int a;                                // variable declaration

cout<<"enter an integer ";            // prompt for user to input data
cin>>a;                               // takes data from user and puts it into
                                         the variable

if(a>0 && a<100)                      // condition for the program to cheak a
                                         number is greater than 0 and less than
                                         100
cout<<"condition met b/t 0 and 100\n";  // if the condition met the message
                                           is sent

else                                    // part that runs if the if condition
                                           fails
cout<<"condition not met"<<endl;        // if the condion doesnt met this
                                           message is sent

return 0;                               // tells the operating system that
                                           program has finished
}
```

Explanation

This program checks whether the entered number is **greater than 0 and less than 100**.

The **logical AND operator (&&)** is used to combine both conditions.
If both conditions are true, the program prints that the number is within the range. Otherwise, it displays a different message.

Task 3: If Statement

Program

```
#include <iostream>                // tells the compiler to include standard
                                      input output library
using namespace std;               // allows us to use cout and cin without
                                      having the need to type std:: for each

int main() {                       // execution of main function beginns from
                                      here
int a;                             // variable declaration

cout<<"enter an integer ";         // prompt for user to input data
cin>>a;                            // takes data from user and puts it in
                                      variable

if(a>0)                            // condition if a is grater than 0
cout<<"the number is positive"<<endl;          // prints message if a is
                                                  greater than 0
else                                           // runs if the if condition
                                                  fails
cout<<"the number is negative"<<endl;          // prints the message if the
                                                  number is smaller than 0

return 0;                                     // tells the operating system
                                                 that program has finished

}
```

Explanation

In this program, the **if statement** checks whether the entered number is **greater than zero**.

If the condition is true, the program prints that the number is positive.
If the condition is false, the program simply ends without printing anything.

Task 4: If-Else Statement

Program

```
#include <iostream>              // tells the compiler to include input output
                                    stream library
using namespace std;             // uses standard namespaces without having to
                                    write std:: for every cin and cout

int main() {                     //the main function from where the execution
                                   starts

int age;                         // variable declaration
cout<<"enter your age";          // asks the user to provide their age
cin>>age;                        // takes the data and puts i into the
                                    variable
if(age>=18)                      //checks if the value in age is 18 or greater
cout<<"you are an edult"<<endl;  // if the condition mets it prints you are an
                                    adult now
else                             // this part runs if the if condition fails
cout<<"you are a minor"<<endl;   // prints you are a minor

return 0;                        // tells the operating system that program
                                    has finishe
```

Explanation

This program asks the user to enter their age.

The **if-else statement** checks whether the age is **greater than or equal to 18**.

- If the condition is true, the program prints that the person is eligible.
- If the condition is false, it prints that the person is not eligible.

Task 5: Else-If Statement

Program

```
#include <iostream>                 // tells the compiler to include
                                       standard input output stream library
using namespace std;                // allows us to use cout and cin without
                                       having the need to typ std:: for each
                                       cin and cout

int main() {                        // main function execution beggins from
                                       here

int a;                              // variable declarartion

cout<<"enter a number";             // displays are request to the user

cin>>a;                             // stores the user input data into
                                       variable
if(a>0)                             // cheaks if the number is greater than 0
cout<<"the number is positive";     // displays thre message if the number is
                                      greater than 0
else if(a<0)                        // if test 1 fails , it cheaks if the
                                      number is less than o
cout<<"the number is negative";     // confirms that the number is negative

else if(a==0)                       // cheaks that if the number is exactly 0
cout<<"the number you entered is zero";        // final confirmation for 0

return 0;                                      // tells the operating system
                                                  that program has finishe


}
```

Explanation

This program determines whether a number is **positive, negative, or zero**.

- If the number is greater than zero, it is positive.
- If the number is less than zero, it is negative.
- If neither condition is true, the number must be zero.

The **else-if statement** allows checking multiple conditions sequentially.